

Network Device & Selection

1. Research two actual network switches available in India (for example, from Cisco, TP-Link, or D-Link) and list their key specifications such as number of ports, speed, and price in a table.

->Here is a comparison of two real network switches currently listed for the Indian market. Prices can vary by seller and promotions.

Switch	Brand	Ports	Port Speed	Type	Key Features	Approx. Price in India
TL-SG108	TP-Link	8 × RJ45	10/100/1000 Mbps (Gigabit)	Unmanaged	Plug-and-play, QoS, IGMP Snooping, fanless metal housing	₹3,862
DGS-108	D-Link	8 × Gigabit Ethernet	1000 Mbps (Gigabit)	Unmanaged	16 Gbps switching capacity, QoS, IGMP Snooping, fanless metal housing	Around ₹2,000–₹2,600

TP-Link's official Indian specification confirms that the TL-SG108 has eight 10/100/1000 Mbps ports, while D-Link's official specifications list eight Gigabit Ethernet ports and a 16 Gbps switching capacity for the DGS-108. ([TP-Link](#))

The current shopping listings I found show the TL-SG108 around ₹3,862, while D-Link 8-port Gigabit models are listed around ₹2,000–₹2,600 depending on the exact model and seller.

Conclusion: Both are 8-port Gigabit switches suitable for small offices, labs, or home networks. They are **unmanaged**, so they are designed for simple plug-and-play connectivity rather than advanced VLAN/routing configuration.

2. Create a device selection matrix comparing at least one switch, one router, and one firewall that could be used for a small startup office (20-30 employees). Include columns for brand/model, key specs, price, and your rating for each device.

Hint: Use online sources like Amazon India, Flipkart, or manufacturer websites for real-world pricing and specs.

->Below is a **device selection matrix for a 20–30 employee startup office**. Prices are recent India-market listings and can change by seller, GST, promotions, and licensing.

Device	Brand / Model	Key specifications	Approx. India price	Rating*
Switch	TP-Link TL-SG2428P	24 × Gigabit PoE+ ports + 4 × Gigabit SFP; 250 W PoE budget; managed/Omada; VLAN and static routing support	₹19,599–₹22,500	4.5/5
Router	TP-Link ER605	5 × Gigabit ports; Multi-WAN; VPN support; Omada SDN management; SPI firewall	₹4,770–₹7,799	4.5/5
Firewall	Fortinet FortiGate 40F	Enterprise firewall; approx. 5 Gbps firewall throughput; UTM/security features; VPN, IPS and web/application controls	₹35,000–₹50,000 with 1-year UTP bundle	4.5/5

*The ratings above are **my non-ranking suitability ratings for a 20–30-person startup**, based on the listed specifications and typical small-office requirements—not customer-review scores.

Why these devices fit a small office

- **TL-SG2428P:** Its 24 Gigabit PoE+ ports can connect PCs, access points, IP phones and cameras, while the four SFP slots provide room for fiber uplinks. TP-Link specifically lists this model for business deployments with a **250 W PoE budget**. ([Flipkart](#))
- **ER605:** Provides five Gigabit interfaces and Multi-WAN/VPN capabilities, making it suitable as the office's Internet gateway. A recent India price listing shows about ₹5,099, while recent Flipkart price-history data places it around ₹5,959. ([Bharathi Systems and Services](#))
- **FortiGate 40F:** Provides dedicated security functions beyond a basic router. An India-focused 2026 pricing guide gives approximately **₹35,000–₹50,000** for hardware plus a 1-year UTP bundle and describes it as suitable for small offices. ([zmtechnologies.com](#))

Example office layout

Internet → FortiGate 40F → ER605/router → TL-SG2428P switch → PCs / Wi-Fi APs / IP phones / printers

For a 20–30 employee office, this combination provides room for wired devices, VLANs, VPN connectivity, PoE devices, and dedicated network security. Note that firewall subscription costs can significantly affect the long-term cost of ownership.

3. Imagine you are setting up the network for a new gaming café with 25 computers and public Wi-Fi. Select one switch, one router, and one firewall model for this scenario and justify your choices in 2-3 lines each.

->**Gaming Café Network Device Selection**

Device	Selected Model	Justification
Switch	TP-Link TL-SG2428P	A 24-port Gigabit managed switch with PoE+ is suitable for connecting gaming PCs, access points, and other network devices. Its VLAN support helps separate gaming PCs from public Wi-Fi traffic.
Router	TP-Link ER605	This 5-port Gigabit router supports Multi-WAN and VPN, providing reliable Internet connectivity and flexibility if the café uses multiple ISP connections. It is also suitable for managing separate network segments.
Firewall	Fortinet FortiGate 40F	The FortiGate 40F provides dedicated firewall, IPS, VPN, and web/application security features. It can help protect the café's gaming network and customer Wi-Fi from unauthorized or malicious traffic.

Suggested topology:

Internet → FortiGate 40F → ER605 → TL-SG2428P → 25 Gaming PCs + Wi-Fi Access Points

This setup also allows separate **Gaming, Staff, and Guest Wi-Fi VLANs**, improving security and network management.

4. Prepare a Bill of Materials (BOM) listing the selected network devices (switch, router, firewall) for your gaming café scenario, including quantity, model, unit price, total price, and a short justification for each item.

->**Bill of Materials (BOM) – Gaming Café Network**

For the previously selected gaming-café setup, the following BOM covers the main **switch, router, and firewall**. Prices are approximate current India-market prices and can change.

No.	Device	Model	Qty.	Unit Price	Total Price	Short Justification
1	Switch	TP-Link TL-SG2428P	1	₹18,671	₹18,671	Provides 24 Gigabit PoE+ ports plus 4 Gigabit SFP slots. Suitable for the café's wired PCs and future access points. (TP-Link)
2	Router	TP-Link ER605 V2	1	₹5,099	₹5,099	Provides 5 Gigabit ports, Multi-WAN capability and VPN support, making it suitable as the café's Internet gateway. (Bharathi Systems and Services)
3	Firewall	Fortinet FortiGate 40F	1	~₹35,000–₹50,000*	~₹35,000–₹50,000	Provides dedicated network-security functions such as firewalling, IPS and VPN protection for the gaming and guest networks.

Estimated BOM Cost

Switch: ₹18,671

Router: ₹5,099

Firewall: ~₹35,000–₹50,000

Estimated total: ₹58,770–₹73,770

*The FortiGate price varies substantially depending on whether security/support subscriptions are included, so it should be confirmed with an Indian Fortinet reseller before purchase.

Network layout:

Internet → FortiGate 40F → ER605 → TL-SG2428P → 25 Gaming PCs + Wi-Fi Access Points

The TL-SG2428P has enough Gigabit ports for a 25-PC café when the router/firewall uplinks and Wi-Fi devices are planned appropriately, with its extra SFP ports providing room for future expansion. ([TP-Link](#))

5. Perform a cost-benefit analysis for your chosen devices: explain in 3-4 lines why you selected these models over cheaper or more expensive alternatives, considering both features and budget.

->**Cost-Benefit Analysis**

- **TP-Link TL-SG2428P:** Chosen because its 24 Gigabit PoE+ ports, VLAN support, and SFP uplinks provide enough capacity for 25 gaming PCs and future expansion without requiring an expensive enterprise switch.
- **TP-Link ER605:** Selected because it offers Gigabit connectivity, Multi-WAN, and VPN features at a relatively low cost, making it more practical for a small café than a higher-priced enterprise router.
- **Fortinet FortiGate 40F:** Although more expensive than a basic router/firewall, it provides dedicated security features such as IPS, VPN, and application/web controls, which are valuable when many customers use the network.
- **Overall:** The combination balances **performance, security, scalability, and budget;** cheaper devices could reduce upfront cost but may lack the management and security features needed for a public gaming-café network.